# Similarity and analogical reasoning: a synthesis

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The ability to perceive similarities and analogies is one of the most fundamental aspects of human cognition. It is crucial for recognition, classification, and learning, and it plays an important role in scientific discovery and creativity. In recent years, similarity and analogy have received increasing attention from cognitive scientists. This growth of interest is related to the realization that human reasoning does not always operate on the basis of content-free general inference rules but, rather, is often tied to particular bodies of knowledge and is greatly influenced by the context in which it occurs. In a reasoning system of this kind, learning does not get accomplished by merely adding new facts and applying the same inference rules to them. Rather, successful learning often depends on the ability to identify the most relevant bodies of knowledge that already exist in memory so that this knowledge can be used as the starting point for learning something new.

All the chapters in this volume have something to say about the role that similarity and analogy play in this process, although they often do so from radically different points of view. This is not surprising since the contributors of this volume come from related but nevertheless distinct disciplines. For some, the focus is on the psychological processes involved in reasoning by similarity and analogy and on how these processes develop. Others are interested in solving the computational problems involved in simulating analogical processing in problem-solving situations. Finally, some researchers, particularly those interested in educational issues, are concerned about the conditions that promote the application of analogical reasoning.

We value this diversity because we believe that an understanding of the role that similarity and analogy play in cognition is most likely to come about by integrating knowledge from diverse specialized do-



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mains. Beyond the interdisciplinary character of the contributions to this volume, however, the differences we find in the treatment of similarity and analogy are also representative of the state of the art in cognitive science today. Because similarity judgments and analogical reasoning are based on people's representation of entities, even the most superficial treatment of these topics requires some assumptions about how knowledge is represented and how these representations change. Although some authors deal with the question of knowledge representation more explicitly than others, this is an issue that runs through most of the chapters in this volume, coloring the treatment of similarity and analogy and accounting for a lot of the diversity we observe.

The purpose of this synthesis is to provide a roadmap to the book, making connections, pointing out similarities and differences, integrating some of the material, and raising some questions that need to be pursued in future research. In an attempt to structure this introduction to correspond roughly to the major groupings of the chapters, we shall discuss first issues having to do with similarity and the structure of concepts; we shall then turn to the nature of analogical reasoning; and finally we shall discuss the implications of existing theories of similarity and analogy for learning and instruction.

## Similarity and the structure of concepts

Two important issues pertaining to similarity emerge from the collection as a whole, and particularly from the chapters in Part I, Similarity and the Structure of Concepts. The first of these concerns the question of what kinds of similarity there are. Is there only one kind, or are there more? If there is more than one kind of similarity, what distinguishes them? The second issue has to do with the relation between conceptual systems and similarity. Here, the key questions are whether perceived similarity determines the nature of our concepts, whether the nature of our concepts determines what we perceive as similar, or whether there is a mutual interdependence between conceptual structures and perceived similarity.

### *Kinds of similarity*

On the question of how many kinds of similarity there are, two basic approaches can be found – approaches that are by no means mutually exclusive. The most widespread of these is based on the explicit or implicit assumption, held by many of the contributors, that there is a

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fundamental difference between something that might be called *surface* similarity and *deep* similarity. The second approach, which is discussed explicitly only by Linda Smith, is to distinguish between *global* and *dimensional* similarity. This distinction essentially focuses on the fact that one can perceive two things as being similar in terms of some holistic perception of them or, in a more constrained way, by viewing them as being similar only along certain discriminable dimensions. In either case, according to Smith, the proper theoretical account has to be in terms of a complex relational system of similarity kinds, rather than in terms of some idealized "pure" similarity.

The contact between the notion of dimensional similarity and the more widespread distinction between surface and deep similarity is that the latter distinction, insofar as it can be clarified sufficiently to be theoretically useful, appears to apply readily to dimensional similarity but to bear no relation to the notion of global similarity. If two things just seem to be similar in general, without reference to any particular dimensions, then the question of whether this perceived similarity is based on surface or deep aspects of the objects does not arise. It arises only when one comes to consider the respects in which two things are similar, for then one can argue that they are similar either in terms of readily accessible dimensions that are apparent on the "surface" or in terms of "deeper" underlying properties that are not.

A clear commitment to the surface/deep distinction can be found in the chapter by Lance Rips. According to Rips, perceptual (i.e, surface) similarity is cognitively primitive and well defined and therefore can be used as a construct to explain other psychological functions, such as categorization. Similarity based on underlying properties is, however, too vague to be useful for such purposes.

In contrast to Rips, Smith is less comfortable with the idea that perceptual similarity is well defined. She argues that there are major shifts with development in the way in which perceptual similarity is distinguished. Early in development children understand the similarities in the way objects look only in terms of global, syncretic resemblances. With development, the relational similarity system becomes differentiated into distinct and interrelated kinds of similarity based on dimensions such as part identify, color, and size, the relations of greater-than, less-than, and so on.

Yet another basis for distinguishing different kinds of similarity emerges from Dedre Gentner's chapter. Gentner makes a distinction between object attributes (one-place predicates characterizing simple descriptions of objects) and relations (two-or-more-place predicates

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characterizing relations). She then goes on to argue that surface similarity is based on shared object attributes, whereas structural similarity is similarity at the level of relational structure.

Finally, Stella Vosniadou argues that information that may be considered as object attributes can be quite difficult to access (e.g., the spherical shape of the earth, the solidity of the moon, etc.) whereas some relations (e.g., movement) can be readily accessible. Vosniadou proposes using the term *salient similarity* instead of *surface similarity* to refer to similarity grounded in easily retrievable aspects of representations, regardless of whether or not they represent descriptive or relational, perceptual or conceptual properties. According to this account, the surface/deep similarity distinction is a dynamic one because in the process of knowledge acquisition the properties of people's underlying representations, which are salient and therefore easy to retrieve, may change.

Douglas Medin and Andrew Ortony also focus on how the distinction between surface and deep similarity might be operationalized, and on what the connection between deep and surface similarity might be. They present the distinction in terms of knowledge representations, suggesting that surface similarities can be thought of as those based on readily accessible components of concepts (of the kind that people describe when asked to list the properties of objects), and deep similarity as similarity with respect to more central, core properties of concepts. One of their main claims is that there is a nonarbitrary, casual relation between surface and central properties, such that the latter constrain and sometimes even generate the former. Medin and Ortony believe that organisms are sensitive to just those kinds of surface similarities that lead them toward the more central properties, a point also made by Ann Brown.

According to Brown, infants come predisposed to seek casual explanations for events (particularly with respect to physical and personal casuality), and these predispositions constrain what is selected from the range of available perceptual attributes to form categories. Children seem to be sensitive to certain perceptual properties of objects that have the possibility of leading to the deeper, more central properties. For example, infants pay particular attention to the movement of objects, noticing very rapidly that some objects need to be pushed or pulled in order to move, whereas other objects do not. This early sensitivity to movement leads infants to differentiate animate from inanimate objects and to continue further differentiating these two categories into finer and finer distinctions as their knowledge of biology and physics increases.

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### *Similarity and conceptual structure*

Because similarity judgments are based on people's representations of entities, a theory of similarity has to be concerned with the nature of these representations and how they change in the process of knowledge acquisition. A number of contributors address this issue quite directly. Lawrence Barsalou argues that there is great variability in the concepts that represent categories and that the resulting instability of concepts affects both within-category and across-category similarity. He argues that careful analyses of how individuals represent specific concepts are needed before one can understand how concepts in general are structured.

An example of such an analysis is provided by Ryszard Michalski. Michalski agrees that concepts have flexible meanings and offers some ideas as to how such flexibility might be achieved. He proposes a two-tiered representation for concepts, according to which the concept meaning is distributed between a base concept representation and an inferential concept interpretation. The base representation contains the "core" of the concept's meaning (the easily explainable concept meaning). Inference processes of all sorts (deductive, inductive, and analogical) operate to relate observations to this base concept representation. Michalski's proposal brings up the question of what the criteria are for distinguishing between the base concept representation and the inferential component, a challenge that is raised, although not resolved, by Gerald DeJong.

An important theoretical issue concerns the question of whether perceived similarity helps shape the nature of concepts or vice versa. A strong denial of the first of these possibilities, that perceived similarity determines the nature of our concepts, is presented in the chapter by Rips. There it is argued that similarity cannot explain how people make category judgments and that resemblance models of categorization (e.g., exemplar and prototype models) may work well for artificial categories but not for richer categories that people know a lot about. Rips presents the results of several experiments suggesting that it is possible to alter similarity judgments without affecting category performance and to alter category judgments without affecting similarity. He concludes that category decisions are better viewed as being based on inferences to the best explanation rather than on similarity of putative members to category prototypes.

Judgments of category membership clearly are pivotal in the creation and maintenance of concepts. The suggestion by Rips that such judgments are perhaps not grounded in similarity is something of a

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radical departure from the prevailing view in cognitive psychology. If similarity is not a crucial component in category judgments, one might wonder whether there are any general psychological processes in which it plays an important role. In the chapter by Edward Smith and Daniel Osherson, an important class of such processes is identified, and a provocative analysis of the role of similarity in decision making is proposed. Thus one might view this chapter by Smith and Osherson as going some way toward restoring confidence in the potential that theories of similarity might have as explanatory mechanisms in important psychological processes. These authors propose a model of decision making according to which a person estimates the probability that an object belongs to a class by relying on the similarity that the object bears to the prototype of the class (e.g., they judge the probability that "Linda" is "a feminist bank teller" by computing the similarity between Linda and their prototype of a feminist bank teller). Smith and Osherson argue that such a model can explain why people often violate certain principles of probability and statistical theory (such as the conjunction rule and Bayes's theorem) as uncovered by Kahneman and Tversky (e.g., 1972, 1973; Tversky & Kahneman, 1982, 1983).

Comparing what might be considered the rather pessimistic view of the significance of similarity as an explanatory construct presented by Rips with the relatively optimistic view presented by Smith and Osherson leads to the possibility that the truth lies somewhere between the two. This is the position that Medin and Ortony take. They argue that similarity may not be as powerful as Smith and Osherson want it to be, although it may play a greater role in categorization than Rips argues for. They acknowledge that the results of Rip's experiments are quite damaging to a theory of categorization that is restricted to similarity in terms only of surface properties, but they see no reason why category judgments cannot in principle be made with respect to similarity of more central aspects of concepts.

## Analogical reasoning

There is general agreement that analogical reasoning involves the transfer of relational information from a domain that already exists in memory (usually referred to as the *source* or *base* domain) to the domain to be explained (referred to as the *target* domain). Similarity is implicated in this process because a successful, useful analogy depends upon there being some sort of similarity between the source domain and the target domain and because the perception of simi-

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larity is likely to play a major role in some of the key processes associated with analogical reasoning.

Three such processes appear to be critical, and discussions of one or more of them surface in virtually every chapter in Parts II and III. These processes are, first, gaining access to an (appropriate) analog; second, mapping some part of the information associated with that analog onto the target domain; and third, the side effects of analogical reasoning in terms of the production of more general rules and representations. In this section, we shall discuss these issues in greater detail while indicating how they relate to a fourth general issue, namely, one having to do with the relation between two distinct kinds of analogies.

The two kinds of analogies we have in mind are those in which the analogically related items are drawn from conceptually different or remote domains and those in which the items are drawn from the same domain, or at least from conceptually very close domains. The first kind might be called *between-domain* (or metaphorical) analogies, and the second we shall refer to as *within-domain* (or literal) analogies. Some of the chapters (e.g., John Anderson and Ross Thompson; Brian Ross) deal exclusively with within-domain analogies, focusing on the use of old problems to solve new ones in the same domain. Others (e.g., Keith Holyoak and Paul Thagard; Gentner; Philip Johnson-Laird) concentrate primarily on between-domain analogical reasoning. We think it is important not to lose sight of the distinction between the two kinds of analogies, because it is possible that somewhat different processes might be involved in the two cases. A discussion of some of these issues can be found in the chapter by Vosniadou.

### *Similarity and the problem of analogical access*

One major way in which similarity is implicated in analogical reasoning is that a theory of analogical reasoning must explain how an analog is retrieved in the first place. In order to access an appropriate analog when constructing an analogy, people must perceive some similarity between the target domain and a source domain. It appears relatively easy to construct a plausible model of the access mechanism in the case of within-domain analogies, which usually exhibit some similarity between target and source domains with respect to both their surface properties and their relational structure. Many empirical findings show that the likelihood of retrieving an analog increases when the surface similarity (simi-

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larity in simple, descriptive properties) between the source and the target is increased (Gentner & Landers, 1985; Holyoak & Koh, 1986; Ross, 1984). Understanding exactly what role surface similarity plays in accessing an analog is complicated, however, by equivocation over exactly what surface similarity is.

The real challenge about access, however, arises in cases where the source domain is remote from the target domain. This is because the likelihood of similarity in surface properties triggering access decreases as the remoteness of the domains increases. One solution for how to account for access when the source domain is very remote from the target domain might be *plan-based indexing*. This is the view that, when a problem cannot be solved with reference to an existing general rule or schema, people start searching for an analog that can provide a solution to the problem that satisfies the problem's goals without violating the target's solution constraints. This proposal was originally suggested by Carbonell (1983), and a modified version of it is utilized in the "processes of induction" (PI), the computer model described by Keith Holyoak and Paul Thagard. In PI there is no explicit search for an analog. Instead, possible analogs can be retrieved as a side effect of the general rule, and concept activation can be generated in the context of the problem-solving activity.

Although plans and goals may play an important role in access, there is doubt that they can account for access completely. Empirical work in analogical problem solving has shown that both adults and children have great difficulty retrieving a remote analog that can satisfy the target problem's goals, even in experimental situations in which such a potential analog has just been provided (e.g., Gick & Holyoak, 1980, 1983). The problem often lies in the fact that neither children nor adults represent the problems at a level abstract (content-free) enough for the plan-based indexing solution to succeed. Furthermore, when they do, they sometimes have access to a specific higher-order schema or concept, which makes the application of analogical reasoning unnecessary. If a specific general schema for solving a problem already exists in memory, why not use that schema instead of reasoning by analogy? This issue is discussed by DeJong in his criticism of the various computational approaches to analogical reasoning.

Another solution is proposed by Vosniadou, who argues that sometimes access to an analog for a between-domain analogy can be based on the identification of some *salient* similarity in the properties of the source and the target. Unlike surface similarity, salient similarity is defined only with respect to the status that it has with respect to

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people's underlying representations and not on the basis of whether it represents simple, descriptive properties.

One interesting finding is that for between-domain analogies access appears to be less of a problem for adults and experts than for children and novices (Brown, this volume; Gentner, this volume; Novick, 1988). What is it about development and knowledge acquisition that allows easier access to analogs from remote domains? One possibility is that adults and experts are more likely to notice and use similarity with respect to relational as opposed to nonrelational properties – that there is what Gentner calls, an "attribute-to-relations shift" with development and with the acquisition of knowledge. Another possibility is that the ability to notice and use similarity in relations is available to children and novices but that children's and novices' knowledge representations do not contain the relational information that is needed when it is needed. This is the position taken by Brown. Yet a third possibility, not inconsistent with the last one, is that access is determined by salient similarity in the properties of the source and target, but what is *salient* similarity may change as people's conceptual representations become reorganized with the acquisition of knowledge (as discussed by Vosniadou). As a result, what may be considered deep properties in the conceptual representations of novices may come to be salient and thus easily accessible to experts (see Carey, 1985; Chi, Feltovich, & Glaser, 1981; Vosniadou & Brewer, 1987).

Research on analogical access has important implications for training and instruction, some of which will be discussed in greater detail later. However, two illustrations of this are worth mentioning now. Brian Ross's suggestion to enhance surface similarity among problems of the same problem type in order to increase the probability that they are placed in the same general category has its source in the finding that surface similarity is an important determinant of access. Similarly, the proposal in the chapter by John Bransford and his collaborators that new material be presented in a problem-solving format is supported by the plan-based indexing view of memory retrieval. In other words, information stored as a problem solution may be more likely to be accessed and used in similar problem-solving situations.

### *Analogical mapping*

Analogical mapping involves setting up correspondences between properties in the two domains and transferring a relational structure

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that embodies some of the relations between these properties. Curiously, however, none of the contributors discusses the conditions under which mapping is required. In many instances of analogy comprehension, transfer of the relational structure is not needed because the relational structure already exists in the target domain.

Analogical reasoning and attendant subprocesses such as access and mapping involve the dynamic interplay of many interrelated systems. For this reason, a number of different computational approaches to analogical reasoning have been developed. Relative to the more traditional techniques of experimental psychology, computer modeling and artificial intelligence techniques provide powerful alternative ways of exploring and testing ideas about analogical reasoning. It should be noted, however, that the adoption of computer modeling techniques in no way eliminates the need for and usefulness of traditional experimental techniques. The present collection includes four different computational approaches to analogical reasoning, or aspects of it, but in almost all cases these models are influenced by experimental data and suggest further experiments. This interaction between computer modeling and psychological experiments is characteristic of the cognitive science approach to dealing with complex information-processing problems. Although there is no doubt that this is a positive development, care needs to be exercised in distinguishing descriptions of computer programs designed to model particular processes from the underlying theories of those processes.

An important difference of opinion in the area of analogical mapping, one discussed in some detail by Philip Johnson-Laird, is whether the emphasis is placed on structural as opposed to pragmatic factors (i.e., ones based on the reasoner's plans and goals). An example of the structure-mapping approach is provided by Gentner. She proposes a structure-mapping process that relies on structural/relational commonalities as opposed to specific prior content and that is not influenced by the system's problem-solving goals, except to the extent that these goals affect the current representation of the domain. In her chapter, Gentner describes how a theory of this kind can be implemented in a computer model that searches for potential local and global matches between the source and target systems, evaluates them on the basis of the depth of the relational match, and selects the deepest, the most systematic, and the most consistent mappable structure.

In contrast to this structure-mapping approach, a more pragmatic approach is possible. For example, Holyoak and Thagard describe a computer model called PI. The basic design of PI involves a bidirectional search in which rules become active as a result of both a forward

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search from the problem situation to possible actions and a backward search from the goal to possible actions and preconditions that could achieve it (the subgoal transfer mechanism). Analogical mapping is used when an activated source analog provides a more efficient solution procedure than a known rule. If a source analog is found, the mapping process (a) maps identical relations, (b) postulates that the relational goals of the two problems are analogous and proceeds from there to map the corresponding arguments of these relations, and (c) hypothesizes an analogical solution to the problem. In this model, the selection of a source analog is a side effect of rule activity directed toward solving the target problem. Holyoak and Thagard consider this to be an advantage over most systems that evade serious examination of the question of when analogies are invoked. PI does not base its selection of source elements to be transferred on structural criteria. Rather, the subgoal transfer mechanism insures that what are transferred are applicable elements of the source problem with functional relevance to a solution.

We have already emphasized the need for a distinction between within-domain analogies and between-domain analogies. Both the Gentner model and Holyoak and Thagard's model are computer models that attempt to handle between-domain analogies, the former by design and the latter by side effect. Within-domain models tend to focus on problem-solving tasks in which the problems to be solved are all of the same general kind (e.g., problems in computer programming or statistics). John Anderson and Ross Thompson describe a computer model (known as PUPS) designed to handle within-domain analogies of this kind, using a production system architecture. The key to their approach lies in the structure of the knowledge representation that they employ. Knowledge is organized in schema-like structures consisting of function and form slots. Analogy is a mechanism for achieving form–function relationships in domains where skills for doing so have not yet been acquired. Analogy does this by finding a source domain that achieves an analogous function (or form) that has the corresponding form (or function) filled. This is achieved by looking for a source in which the first element in the function slot is identical to the corresponding element in the target. The remaining elements are put into correspondence. The system uses various inductive inferences in the mapping.

The computational approaches that we have discussed so far all lie within the symbolic representation paradigm. Knowledge structures and rules for operating on them are explicitly represented in symbolic form. During the last ten years or so, an alternative computational

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approach known as *connectionism* has become increasingly influential. Connectionism eschews the idea of explicit symbolic representations. It is based on the idea that representations are not localized and symbolically represented but are emergent properties of memory and learning systems that are composed of a very large number of computationally very simple (neuronlike) elements. Such systems have been developed to perform a wide array of relatively low-level tasks (such as letter and word recognition), but they have not yet reached the point of being able to deal with high-level, complex tasks such as analogical reasoning. Nevertheless, it is important to consider how such parallel distributed processing systems might, in principle, be able to approach such problems. A speculative account of how this might be done can be found in the chapter by David Rumelhart. Rumelhart describes such a distributed memory system in which collections of elements, which he calls microfeatures, could represent both the physical characteristics of a situation and its more relational aspects. Depending on how familiar it is, when an input enters the memory system it is either recognized and assimilated to stored patterns, or rejected by the system (at this point the activity of the memory system is shut down). Analogical reasoning would occur in situations in which the information stored in memory provided no close match.

It is clear that accounts of the computational approaches of the kind we have been discussing represent powerful ways of exploring some of the complex issues surrounding the problem of analogical reasoning and mapping. However, they raise some problems that are not easy to resolve. One of these concerns the question of how to choose among competing accounts. The difficulty here lies in the fact that most implemented models have unavoidable limitations of scope that make it almost impossible to undertake fair and objective comparisons of performance. Another, emphasized by Johnson-Laird, concerns the possibility that computational models of analogy are in principle limited to cases of what he calls *expository* analogies, but that *profound* analogies, which he believes involve genuine creativity, are computationally intractable.

### *Analogy and generalization*

For most researchers who work in the area of problem solving, analogical reasoning (mainly within-domain analogical reasoning) is a mechanism used primarily by novices who lack general rules or schemata. There is reason to believe that young children (who also lack general rules and expertise in domains) use analogical reasoning as

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a primary problem-solving and knowledge acquisition mechanism. Given that novices often use analogical reasoning in problem solving, one important question with serious implications for learning is whether the successful application of analogical reasoning helps novices learn general rules and build schemata they lack. Many researchers believe so, although the question of how and when such generalizations are made is far from clear (as discussed by Allan Collins and Mark Burstein in the final chapter).

According to Anderson and Thompson, analogy is used only at the beginning stages of problem solving. In PUPS, every time an analogy is used, the abstract implicational structure is extracted and a rule is formed. This prevents the system from having to use an analogy a second time. Overgeneralization is prevented by adding heuristic constraints based on empirical cases.

For Holyoak and Thagard, analogy is seen as a trigger for induction. A generalized rule is induced after the analogy is used to form an abstract problem schema. It could also be argued (as it is by Ross, for example) that the process of analogical reminding itself may force certain very conservative generalizations, which nevertheless can lead students to learn about problem types. Such a process may be one means by which novices go from superficial similarities to structural similarities, given that the two are related. The use of surface or salient similarity as a means for discovering, transferring, and generalizing a casual mechanism by young children is also discussed by Brown and by Vosniadou in the context of discussing the developmental literature on analogical reasoning.

Finally, Rand Spiro and his colleagues in considering the use of explicit, instructional, between-domain analogies in ill-structured domains give a number of examples of people's tendency to overgeneralize on the basis of single analogies. The problem here is that, whereas analogies help novices gain a preliminary grasp of difficult concepts, they may later become impediments to fuller or more correct understandings.

## Implications for instruction

A major challenge facing research on learning and instruction is the question of how to liberate "inert" knowledge. The issue here is not that the problem solver lacks the knowledge to solve the problem but rather that the problem solver cannot recruit that knowledge as and when it is needed. This problem is a major theme in Part III. Starting with the premise that knowledge is often isolated and context-bound and that

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this makes the noticing of similarity and the spontaneous application of analogical reasoning difficult, some researchers (e.g., Bransford et al.; Brown) focus on how the knowledge base can become more flexible and discuss certain instructional techniques for making it so. Others (e.g., Ross; Vosniadou; Spiro et al.) think that reasoning by similarity and analogy itself can make the knowledge base more flexible and facilitate the learning of general rules and the acquisition of new schemata (see also the discussion above on analogy and generalization).

Bransford and his colleagues think that the problem of inert knowledge is related to the way people are taught. Often, instruction involves teaching facts that can be recalled but cannot be used in new situations. Experiments show that instructional procedures that result in learning (in the sense of being able to recall relevant information) provide no guarantee that people will use this information later on. Bransford et al. suggest that this problem can be solved by teaching people in problem-oriented learning environments rather than by the mere presentation of factual information.

Bransford and his colleagues also discuss how instruction can facilitate the noticing of relevant features of problem situations and how new advances in instructional technology make it possible to teach in new and more effective ways. For example, videotapes can be used to create contrasts that highlight the similarities and differences between concepts. It is also important to encourage students to take multiple perspectives on the same problem and to understand how different concepts can illuminate it (a point also made by Spiro, et al., as well as by Brown).

Like Bransford and his collaborators, Ross emphasizes the context dependence and isolation of knowledge. Ross argues that people have difficulty in transferring information not only when what has been learned needs to be transferred across remote domains, but even in the same domain. Although the phenomenon of highly contextualized knowledge may be the result of lack of expertise, much of what we believed to be accomplished by abstract processing mechanisms may be accomplished by more specific mechanisms based on highly similar materials (a point also made by Rumelhart). Surface similarity may often enhance the probability of people using analogical reminding to notice that problems belong to the same problem type.

Vosniadou suggests that similarity in salient properties between the source and target domains can lead to the discovery of a productive analogy, that is, an analogy that can produce *new* knowledge about the structure of the target system. In this respect, analogy can become an important mechanism in the acquisition of new knowledge. Ac-

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cording to Vosniadou, children use analogical reasoning to acquire new knowledge about a domain just like adults do. What develops is not the analogical mechanism itself but the conceptual system upon which this mechanism operates.

In her chapter, Brown approaches the problem of inert knowledge by arguing for the importance of functional flexibility. A potential solution to a problem is often not seen although it is in plain sight, because it is embedded in a familiar, but currently irrelevant, context. This suggests that it is important to experiment with objects in many different ways, to look at them from different points of view. Such activities are likely to free objects from specific roles, thus rendering them more useful as tools for more varied applications. Brown's experimental findings with young children show that experiences that enhance flexibility lead to increases in transfer, whereas the habitual use of an object to perform a specific task produces a lower rate of transfer.

In considering the role of analogical reasoning in instruction, little attention has been paid to the process of understanding an explanatory analogy explicitly presented to facilitate knowledge acquisition in an unfamiliar domain. By far the lion's share of the theoretical attention has been devoted to the question of analogy creation, although Spiro and his colleagues discuss a number of issues related to the cognitive consequences of analogy comprehension. Because of their concern over the potential that analogies have for leading to false or oversimplified representations, Spiro and his collaborators focus on instructional strategies that might avoid such undesirable consequences. They recommend using explicit analogies to facilitate the process of knowledge acquisition in ill-structured domains where teaching rules is not possible (even if it were desirable) and where there is considerable conceptual load. Their technique involves the use of integrated multiple analogies to counteract missing or misleading information that a single analogy might contain.

It appears that analogical reasoning can fulfill a number of useful functions to facilitate learning. What is still lacking, however, is a systematic treatment of the differences and similarities among these various functions and of the unique problems and/or advantages each one presents.

## Conclusion

The chapters in this volume advance our understanding of similarity and analogy on a number of different fronts. On the issue of similarity,

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they make it clear that there is a complex relational system of similarity kinds. This similarity system is not static but changes with development and the acquisition of knowledge. Furthermore, because similarity judgments are based on people's representations of entities, any developments in understanding the structure of the similarity system will be related to developments in understanding how concepts are represented and how this conceptual structure changes with knowledge acquisition.

One important issue that has emerged from the contributions to this volume is that of the distinction between surface similarity and deep similarity. Several contributors believe that there are systematic causal relations between the two and that similarity in the surface properties can be used as the basis from which to infer similarity in deeper (or more central) properties. Further progress in this area, however, depends crucially on a better understanding of what constitutes more surface versus deep similarity and how this distinction is related to conceptual structure.

The question of analogical mapping has been approached from a number of different points of view (e.g., structure-mapping, pragmatic). Also, various computational approaches to analogical mapping have been presented. It has become clear that, from the production point of view, analogical reasoning is a common mechanism used by novices and children to solve problems and answer questions. However, experts use analogical reasoning too. When they do, the main problem becomes one of access rather than one of mapping.

Explicit analogies can also be used for the purpose of instruction. Here, the question of access becomes irrelevant because the analog is given. Now, the question of mapping comes into focus, particularly in situations where the person who has to understand the analogy has a relatively impoverished structure for the target domain. In this situation, mapping is likely to result in a richer representation. It would be a mistake to suppose, however, that the communicative and instructional force of an analogy is restricted to the creation of new cognitive structures. Analogy may sometimes serve the purpose of highlighting what is already known. This highlighting could be effected either through changing the salience of the properties of the target representation or by providing a metaphorical way of labeling relations that are difficult to express using literal language.

Finally, the views presented in this volume have some important practical applications. For example, it appears that analogical learning and transfer of knowledge are facilitated when people are taught in problem-oriented learning environments as opposed to being taught

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facts; when the surface similarity between what is taught and the situations in which it needs to be transferred is enhanced; and when students are encouraged to take many perspectives and to look at objects from different points of view.

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